

EXPERIMENT 10

IMPLEMENTATION OF 3DES

AIM

To implement a simple Triple DES (3DES) encryption and decryption using three keys in C.

ALGORITHM

1. Start the program.
2. Read the plaintext.
3. Read three secret keys.
4. Encrypt the plaintext using Key1, Key2, and Key3.
5. Decrypt using Key3, Key2, and Key1.
6. Display the encrypted and decrypted text.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char text[100], k1, k2, k3;
    unsigned char enc[100];
    int i, len;

    printf("Enter Plain Text: ");
    fgets(text, sizeof(text), stdin);
    text[strcspn(text, "\n")] = '\0';

    printf("Enter Key 1: ");
    scanf(" %c", &k1);

    printf("Enter Key 2: ");
    scanf(" %c", &k2);

    printf("Enter Key 3: ");
    scanf(" %c", &k3);

    len = strlen(text);
```

```

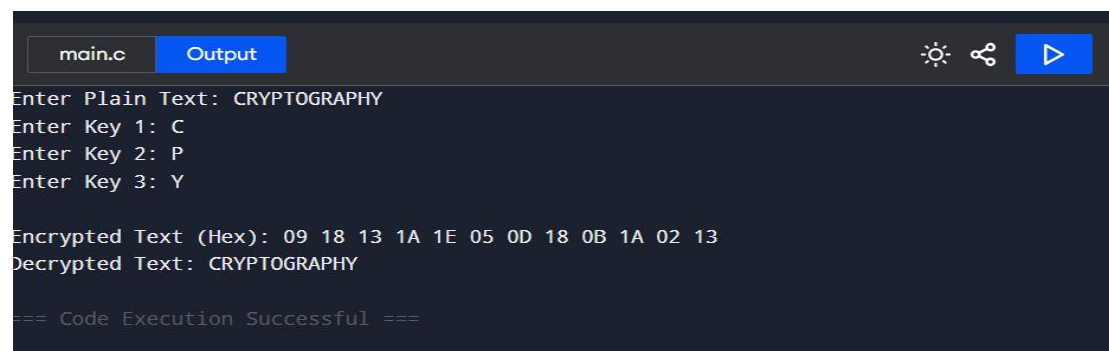
printf("\nEncrypted Text (Hex): ");
for(i = 0; i < len; i++)
{
    enc[i] = ((text[i] ^ k1) ^ k2) ^ k3;
    printf("%02X ", enc[i]);
}

printf("\nDecrypted Text: ");
for(i = 0; i < len; i++)
{
    printf("%c", ((enc[i] ^ k3) ^ k2) ^ k1);
}

return 0;
}

```

OUTPUT



The screenshot shows a code editor with a file named 'main.c' and an 'Output' tab. The output text is as follows:

```

Enter Plain Text: CRYPTOGRAPHY
Enter Key 1: C
Enter Key 2: P
Enter Key 3: Y

Encrypted Text (Hex): 09 18 13 1A 1E 05 0D 18 0B 1A 02 13
Decrypted Text: CRYPTOGRAPHY

=== Code Execution Successful ===

```

RESULT

The plaintext was successfully encrypted and decrypted using a simple Triple DES (3DES) simulation.